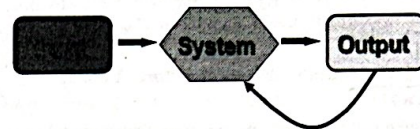


Scientific Experiments and Error Controlling

1. Planning of Experiments
2. Causes of Experimental Errors
3. Fundamental Principles of Error Controlling
4. Features and Basic Requirements of Field Experiments

1

- Experimentation allows an investigator to find what happens to the output or response when the settings of the input variables in a system are purposely changed.



- Experimentation is used to understand and improve a system (product / process).

2

Four Eras in the History of Exp.Des.

- The **agricultural** origins, 1908 – 1940s
 - Field Trials at Rothamsted (1843 – 1846)
 - W.S. Gossett and the *t*-test (1908)
 - R. A. Fisher & his co-workers
 - Profound impact on agricultural science
 - Factorial designs, ANOVA
- The **first industrial** era, 1951 – late 1970s
 - Box & Wilson, response surfaces
 - Applications in the chemical & process industries
- The **second industrial** era, late 1970s – 1990
 - Quality improvement initiatives in many companies
 - Taguchi and robust parameter design, process robustness
- The **modern** era, beginning circa 1990

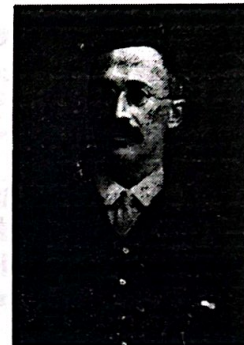
3

William Sealy Gosset (1876-1937)

Gosset's interest in barley cultivation led him to speculate that design of experiments should aim, not only at improving the average yield, but also at breeding varieties whose yield was insensitive (robust) to variation in soil and climate.

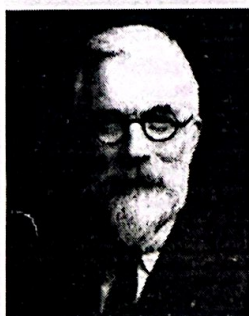
Gosset was a friend of both Karl Pearson and R.A. Fisher, an achievement, for each had a monumental ego and a loathing for the other.

Gosset was a modest man who cut short an admirer with the comment that "Fisher would have discovered it all anyway."

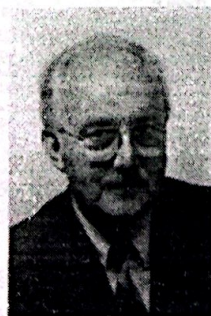


Student in 1908

4



R. A. Fisher (1890 – 1962)



George E. P. Box (1919 – 2013)

5

Basic methods of Scientific Research

- Topic: creativity, possibility, feasibility
- Reference: the basis for topic
- Hypothesis
- Test of hypothesis
- Plan and design of experimentation
 - Principle of "unique difference"
 - Experimental design is an experimental setting to eliminate system errors and control random errors

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1 Planning of Experiments

- **Experimental Indicator:** indication traits that are used to measure the experimental effects
- **Factor:** controllable experimental variables affecting the outcome or output of experiment
- **Level:** various levels or statuses of experimental factors
- **Treatment:** combinations of different levels of factors
- **Experimental effect:** increase or decrease of experimental indicators by experimental factors
- Main effect; interaction effect

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- **Example 1** Objective: saving fuel
 - Fuel consumption
 - Automobile type (large trucks, small car)
 - Driver (novice, driving time below 2 years, driving time above 2 years)
 - Temperature around (0°C, 15°C, 30°C)
 - Humidity (20%, 40%, 60%)

- **Example 2**
Objective: find a suitable fertilization plan
 - Yield
 - Fertilizer type (N, P)
 - Fertilization quantity (10kg/亩, 15kg/亩)
 - Rice variety (V1, V2, V3)

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- Please design an experiment to study the influence on vital capacity by two fitness exercises

- State the experimental indicators, factors, levels and other possible factors having influences on the experimental result

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Experimental Steps:

1. State the objective
2. Choose experimental indicators, response
3. Choose factors, levels, and ranges
4. Choose a design (RCBD, Factorial et al.)
5. Implement the experiment
6. Statistical analysis of experimental data
7. Draw conclusions and make recommendations

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Choose experimental indicators:

- Output or observed result of Experiment
- Measuring Errors make analysis more difficult

Choose experimental factors, levels, ranges

- Arrange comparison
- If only one factor exists, its level is the treatment.

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(Factor 1) Variety	Fertilization (factor 2)		
	N100	N150	N200
V1	Treatment 1: V1 & N100	Treatment 2: V1 & N150	Treatment 3: V1 & N200
V2	Treatment 4: V2 & N100	Treatment 5: V2 & N150	Treatment 6: V2 & N200
V3	Treatment 7: V3 & N100	Treatment 8: V3 & N150	Treatment 9: V3 & N200

Variety (3 levels)	Fertilization (3 levels)	Treatment
V1	N100	Treatment 1: V1 & N100
	N150	Treatment 2: V1 & N150
	N200	Treatment 3: V1 & N200
V2	N100	Treatment 4: V2 & N100
	N150	Treatment 5: V2 & N150
	N200	Treatment 6: V2 & N200
V3	N100	Treatment 7: V3 & N100
	N150	Treatment 8: V3 & N150
	N200	Treatment 9: V3 & N200

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Put the treatments into different experimental units

3 lines repeat			Block I	Block II	Block III
Treatment 3	Treatment 8	Treatment 8	Treatment 3	Treatment 1	Treatment 4
Treatment 1	Treatment 7	Treatment 6	Treatment 6	Treatment 5	Treatment 5
Treatment 5	Treatment 1	Treatment 6	Treatment 5	Treatment 4	Treatment 6
Treatment 5	Treatment 4	Treatment 2	Treatment 4	Treatment 6	Treatment 7
Treatment 6	Treatment 4	Treatment 8	Treatment 8	Treatment 9	Treatment 8
Treatment 9	Treatment 3	Treatment 7	Treatment 1	Treatment 7	Treatment 1
Treatment 2	Treatment 3	Treatment 2	Treatment 9	Treatment 3	Treatment 3
Treatment 9	Treatment 1	Treatment 7	Treatment 7	Treatment 2	Treatment 2
Treatment 9	Treatment 3	Treatment 5	Treatment 2	Treatment 8	Treatment 9



Experimental Unit: the smallest unit of the experiment, any two different units may be arranged by different treatments

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Quantitative Factors: 数量性状

- ☐ Factor taking values over a continuous range.
- ☐ More freedom in choosing the levels.
- ☐ Example: for range 100°C-200°C, choosing 130°C and 160°C for two levels, or 125°C, 150°C, 175°C for three levels.
- ☐ Two levels are adopted for a linear effect, three or more levels for a curvature effect
- ☐ Enough differences among levels are needed, but not too large.

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Levels for factors

Linear effect

Curvature effect

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Qualitative Factors: 质量性状

- ☐ Factor taking on a discrete number of values.
- ☐ Less flexibility in choosing the levels.
- ☐ In choosing factors and levels, cost and practical implementation must be considered.
- ☐ Potentially important factors should be taken into account in the planning of experiment.

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☐ Single-factor Experiment

Simple, easy conducted, one level served as check
Variety trial, fertilize test, pesticide test

☐ Multiple-factor Experiment

More informative and efficient, interaction
Regional trial (Y, L, G, YxL, YxG, LxG, YxLxG)

☐ Comprehensive Experiment

Unbalanced, superior treatment combination
Choose an optimized comprehensive high yield technique suitable for local environment

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Choose Experimental Plan

- ☐ A poor design may capture little information, can't be rectified even if by advanced analysis methods
- ☐ If the experiment is well planned, it is possible to obtain the obvious results without sophisticated analysis.

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Perform the Experiment

- It is necessary to do the preliminary experiment
- Any inconsistent with the planned experiment is need to be recorded

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2 Sources of Experimental Errors

- Example 1: protein content of Nannong 88-48 soybean
 - Twice measure on the same sample: 42.35%, 41.98%
- Example 2: It is known that the true value of the calcium carbonate content of some water sample is 20.7mg/L
 - We investigate the water sample for 11 times repeatedly, the contents of calcium carbonate are 20.99, 20.41, 20.10, 20.00, 20.91, 22.60, 20.90, 20.41, 20.00, 23.00, 22.00

$$y = \mu + \xi$$

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试验误差的来源分类

- Experimental Error: variation between observed value and truth value, variation in the experimental units on which the same treatment is assigned.

Example : Nannong 88-48 seed

Frozen sample:

38.60 , 39.50 , 37.40 , 38.40 , 38.25

Normal sample:

41.20 , 40.80 , 41.50 , 40.60 , 42.35

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□ Non-Random Error:

- **Systematic Error:** the measured values or the computed statistics are consistently higher or lower than the true value or the true parameter in the same situation, indicating the error sizes have obvious orientation

- **Non-Systematic Error:** errors caused by accidental mistakes (such as reading or writing mistakes) made by researcher.

- **Random Error:** even we measure the same object under the same condition repeatedly, and there's no accidental mistakes or systematic errors, there are still some unoriented random errors

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试验误差的来源

Fertilization (Factor 2)				Block I	Block II	Block III
(Factor 1) Variety	N100	N150	N200	Treatment 3	Treatment 1	Treatment 4
V1	Treatment 1: V1 & N100	Treatment 2: V1 & N150	Treatment 3: V1 & N200	Treatment 6	Treatment 5	Treatment 5
V2	Treatment 4: V2 & N100	Treatment 5: V2 & N150	Treatment 6: V2 & N200	Treatment 5	Treatment 4	Treatment 6
V3	Treatment 7: V3 & N100	Treatment 8: V3 & N150	Treatment 9: V3 & N200	Treatment 4	Treatment 6	Treatment 7
				Treatment 8	Treatment 9	Treatment 8
				Treatment 1	Treatment 7	Treatment 1
				Treatment 9	Treatment 3	Treatment 3
				Treatment 7	Treatment 2	Treatment 2
				Treatment 2	Treatment 8	Treatment 9

Sources of Errors:
variety, fertilization, soil
variance, agricultural
operation, pest, climate

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Sources of Experimental Errors:

- Inconsistency of Experimental Materials
 - **Homogeneity of experimental materials**
 - **Animal:** species, strain, age, gender, nest, weight, nutrition, health status, and so on
 - **Sample:** variety, lot number, period of validity, dosage, and so on
 - **Human:** gender, age, level of education, and so on
- **Experimental Management:** experimental operation, fertilizer management, and so on
- Experimental units on which the same treatment is assigned
- Measuring Error

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控制误差的三种方法

3 Fundamental Principles of Error Controlling

□ Three fundamental principles:

1. Replication
2. Randomization
3. Blocking



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Replication

- Make the experimental error be estimable
- Increase the replications to improve the ability of detecting differences between treatments
- Three kinds of meanings
 - Each sample is divided into k parts tested for k times, the arithmetic mean is treated as the observed value (reducing the errors from methods or operations of experiment)
 - Assign each treatment to k experimental units under the same experimental condition (detecting the difference among experimental units by measuring sample for k times)
 - Investigate the same sample for k times (detecting the variation among k times of measures)

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Randomization

- (1) Applied when assigning different treatments to various experimental units
- (2) Arrange the order of treatments during implementation
- (3) The order for measuring dependent variable
- In the process of sampling or allocating units to treatments, it is necessary that every individual in the population has equal chance to be sampled, and every individual in the sample has equal chance to be arranged to any treatment unit
- Unknown variables that may impact the responses in the experiment
- Ensure the validity of the estimates of experimental errors
- Reduce the unwanted influence resulted from subjective judgments in treatment allocation

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Block

- Block (Local Control): a group of homogenous units
- Block can be days, weeks
 - Effective blocking: within-block variation is much smaller than that of between-block
 - Comparing the treatment within the same block
 - Remove the variation between blocks
- Arrange block for what you can control, and randomize for what you can not
 - Randomizing treatments within block

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Randomization, Block

Block I	Block II	Block III
Treatment 3	Treatment 1	Treatment 4
Treatment 6	Treatment 5	Treatment 5
Treatment 5	Treatment 4	Treatment 6
Treatment 4	Treatment 6	Treatment 7
Treatment 8	Treatment 9	Treatment 8
Treatment 1	Treatment 7	Treatment 1
Treatment 9	Treatment 3	Treatment 3
Treatment 7	Treatment 2	Treatment 2
Treatment 2	Treatment 8	Treatment 9



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- Please design an experimentation to study the impact of two kinds of fitness sport on vital capacity
- Experimental Indication: vital capacity
- Factor: fitness sport (sport 1, sport 2, comparison)
- Possible factors that may impact the experimental response: gender, age, sport manner (time, intensity, and so on), measuring errors, cost, and so on

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田间试验

有机体

多变化
的自然环境

4 Features and Basic Requirements of Experiments In Field

□ Features of Experiment in Field:

- (1) The research objects and materials are organism (crops, insects, bugs, animalcule in soil, weed, and so on), the experimental effect is directly detected by the response of organism itself. Variability of organism often results in many sources of experimental errors.
- (2) conducted under open natural environment, the experimental conditions include soil, climate, even some biological conditions like pests. Changeful and long duration, investigating under natural environment in field will result in systematic errors and random errors introduced.

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田间试验误差来源

□ Sources of Errors in Experiments in Field

1. *Inherent differences* between the experimental materials (genetic, non-genetic)
2. *Inconsistency between agricultural operation and management technique* (1) Inconsistencies in soil preparation, fertilization and depth in sowing, (2) Inconsistency between management time in field and operation standard, (3) Inconsistencies between testing time, persons, instruments, and so on.
3. *Differences between outer experimental conditions* (differences between soil, pests, trampling of human and animals, impact of storms, and so on).

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控制田间试验误差的方法

改进方法

□ Methods to Control Errors in Experiments in Field

1. Allocating the homogenous and consistent experimental materials (genotype, growth and development) into the same block.
2. Effort to Keep identical operation and management technique, all the management, investigation and measures, data collection should be carried out in blocks to reduce the possible variation.
3. Control the exterior factors that arouse differences: (1) Choose the experimental field with uniform fertility; (2) Adopt appropriate plot technique; (3) Employ superior experimental design and corresponding statistical analysis.

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□ Selection and improvement of Experimental Field

1. *Uniform fertility in field.* Those fields, where fertilizer or cultivation experiment are once conducted, are not suitable for experiment. Or they can be used after one or several uniform planting.
2. *Choose the fields with same or similar utilization.*
3. *Flat field* is the best choice. All plots within same replication are arranged in the same contour to keep consistent fertility and drainage conditions of field.
4. *Choose appropriate site.* It is best to plant same crop around the experimental field.
5. It is best to *perform blank test* for chosen fields.
6. *Adopt rotation system* for the experimental fields.

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□ Plot techniques for Controlling Soil Differences

■ 1. Area of the experimental plot

In a certain range, the experimental errors decrease as the plot area increases. The accuracy may increase by the increase of plot area, but will decrease as the number of replication decreases. Generally, the area of experimental plot should be in the scope of 6-60m².

■ 2. Shape

Rectangular plot, especially strip-shaped plot, are convenient to smooth soil differences, investigate, record, and conduct field operation. The longer side of plot should be vertical with the direction in which the fertility of field changes the greatest, so that the block direction and the fertility gradient are perpendicular to each other.

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■ 3. The number of replication

The experimental errors decrease as the number of replication increases. The number of replication should be specified according to the required accuracy, experimental soil variation, the amount of experimental materials, the area of experimental field, the size of blocks, and so on.

■ 4. Setting the check (对照区)

Check (CK) should be set to serve as a standard for comparison. The check should be the local improved variety or the most widely used planting techniques.

The purpose of setting check is: (1) to easily investigate and compare treatments in field; (2) to estimate and rectify the soil variation of the experimental field.

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■ 5. Setting the guarding rows

The merits of setting guarding rows: (1) to protect the experimental materials from other factors such as trampling and damages from human and livestock; (2) to prevent the marginal effect, so that treatments are compared correctly.

■ 6. Arrangement for replication (or blocks)

all treatment plots are arranged in a relatively homogeneous field land, which is called block. The number of replication is generally 3 or 4.

■ Complete block: The replication is the same as the block, every block or replication contains a full set of treatments.

■ Incomplete block: A replication is allocated on several blocks, each block only contain partial treatments.

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■ The repeats or blocks can be arranged in one, two, or more rows in field. This depends on the shape and topography of the experimental land, and so on. Particularly, soil differences should be considered. The principle is in effort to keep consistent fertility within same replication or block, while differences between replications are permitted.

■ The positions of plots within replication are arranged in random, to avoid systematic errors, increases the experimental accuracy. Furthermore, it could provides unbiased estimation of errors.

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Common Experimental Design

□ 1. Experimental design that is sequentially arranged

■ Unable to provide unbiased estimation for errors, possibly has systematic errors

1) Contrast design (对比法设计)

■ Used for comparison or demonstration experiments with few varieties. Each variety in trial is directly arranged along the check, so that each plot and the check next to it can be compared directly. All plots within each replication are sequentially arranged

■ Advantage: it is easy to achieve accurate result, and conduct implementation and investigation in field

■ Disadvantage: too many checks, inefficient to use field

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I	1	CK	2	3	CK	4	5	CK	6	7	CK	8
II	7	CK	8	1	CK	2	3	CK	4	5	CK	6
III	5	CK	6	7	CK	8	1	CK	2	3	CK	4

Arrangement of 8 varieties and 3 times repeat in contrast design

Analysis methods:

1. Calculating the ratio in percentage of each variety to its neighboring check
2. Unable to perform significance test

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□ 2) Interval contrast design (间比法设计)

■ Applied at the early stage of breeding experiment when there are too many cultivars in appraisal garden trial, and difficult to arrange random block, but lower accuracy requirement

■ The feature of interval contrast design is that, the first plot and the last plot in a row are both CK.

■ Equal number of treatments between each pair of neighboring checks are arranged, which is generally 4 or 9, and is repeated 2 to 4 times.

■ Advantage: easy to design, convenient to operate, and is able to reduce the marginal effect and growth competition.

■ Disadvantage: the arrangement of the treatments in a block is not randomized. There's possible systematic errors in comparison of varieties. The estimation of experimental errors is biased. What's more, it is unable to perform significance test.

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CK	1	2	3	4	CK	5	6	7	8	CK	9	10	11	12	CK	13	14	15	16	CK
CK	16	15	14	13	CK	12	11	10	9	CK	8	7	6	5	CK	4	3	2	1	CK
CK	1	2	3	4	CK	5	6	7	8	CK	9	10	11	12	CK	13	14	15	16	CK

Analysis methods:

Calculating the ratio in percentage of each variety to the average of its neighboring 2 checks

Unable to perform significance test

Improved interval contrast design: the checks are allocated to different columns, the arrangement of plots corresponding to varieties is randomized

Randomized																				
CK	1	2	3	4	CK	8	9	10	11	12	CK	13	14	15	CK	CK				
CK	15	14	13	CK	12	11	10	9	CK	8	7	6	5	CK	4	3	2	1	CK	CK
CK	CK	1	2	3	4	CK	8	9	10	11	12	CK	13	14	15	CK				

Analysis methods:

The ANOVA can be used to test the significance difference between varieties

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2. Experimental Design in Randomized Arrangement

□1) Complete Random Design

■ Randomly allocate each treatment to the experiment unit (or plot), the number of replication of each treatment can be the same or not. This design is flexible for the arrangement of experimental units, and is applicable both in single or multiple factors trial.

■ Advantage: easy to analyze,

■ Disadvantage: the environmental factor in the experiment should be rather uniform, only suitable for cultivation trial in laboratory, and pots bowl trial in greenhouse.

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3 times repeat

Treatment 3	Treatment 8	Treatment 8
Treatment 1	Treatment 7	Treatment 6
Treatment 5	Treatment 1	Treatment 6
Treatment 5	Treatment 4	Treatment 2
Treatment 6	Treatment 4	Treatment 8
Treatment 9	Treatment 4	Treatment 7
Treatment 2	Treatment 3	Treatment 2
Treatment 9	Treatment 1	Treatment 7
Treatment 9	Treatment 3	Treatment 5

$$y_{ij} = \mu + T_i + e_{ij}$$

分析方法:

采用单因素方差分析方法,
分析处理效应
可对处理效应作显著性检验

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□2) Random Block Design

Block I	Block II	Block III
Treatment 3	Treatment 1	Treatment 4
Treatment 6	Treatment 5	Treatment 5
Treatment 5	Treatment 4	Treatment 6
Treatment 4	Treatment 6	Treatment 7
Treatment 8	Treatment 9	Treatment 8
Treatment 1	Treatment 7	Treatment 1
Treatment 9	Treatment 3	Treatment 3
Treatment 7	Treatment 2	Treatment 2
Treatment 2	Treatment 8	Treatment 9

$$y_{ij} = \mu + T_i + B_j + e_{ij}$$

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□3) Latin Square Design

3×3			4×4				5×5					6×6						7×7						
A	B	C	A	B	C	D	A	B	C	D	E	A	B	C	D	E	F	A	B	C	D	E	F	G
B	C	A	B	C	A	D	A	B	C	D	E	B	C	D	E	F	G	B	C	D	E	F	G	H
C	A	B	C	A	B	E	B	C	D	E	F	C	D	E	F	G	H	C	D	E	F	G	H	I
A	B	C	D	E	F	A	B	C	D	E	F	A	B	C	D	E	F	A	B	C	D	E	F	G
B	C	D	E	F	G	B	C	D	E	F	G	B	C	D	E	F	G	B	C	D	E	F	G	H
C	D	E	F	G	H	C	D	E	F	G	H	C	D	E	F	G	H	C	D	E	F	G	H	I
D	E	F	G	H	I	D	E	F	G	H	I	D	E	F	G	H	I	D	E	F	G	H	I	J
E	F	G	H	I	J	E	F	G	H	I	J	E	F	G	H	I	J	E	F	G	H	I	J	K
F	G	H	I	J	K	F	G	H	I	J	K	F	G	H	I	J	K	F	G	H	I	J	K	L
G	H	I	J	K	L	G	H	I	J	K	L	G	H	I	J	K	L	G	H	I	J	K	L	M
H	I	J	K	L	M	H	I	J	K	L	M	H	I	J	K	L	M	H	I	J	K	L	M	N
I	J	K	L	M	N	I	J	K	L	M	N	I	J	K	L	M	N	I	J	K	L	M	N	O
J	K	L	M	N	O	J	K	L	M	N	O	J	K	L	M	N	O	J	K	L	M	N	O	P
K	L	M	N	O	P	K	L	M	N	O	P	K	L	M	N	O	P	K	L	M	N	O	P	Q
L	M	N	O	P	Q	L	M	N	O	P	Q	L	M	N	O	P	Q	L	M	N	O	P	Q	R
M	N	O	P	Q	R	M	N	O	P	Q	R	M	N	O	P	Q	R	M	N	O	P	Q	R	S
N	O	P	Q	R	S	N	O	P	Q	R	S	N	O	P	Q	R	S	N	O	P	Q	R	S	T
O	P	Q	R	S	T	O	P	Q	R	S	T	O	P	Q	R	S	T	O	P	Q	R	S	T	U
P	Q	R	S	T	U	P	Q	R	S	T	U	P	Q	R	S	T	U	P	Q	R	S	T	U	V
Q	R	S	T	U	V	Q	R	S	T	U	V	Q	R	S	T	U	V	Q	R	S	T	U	V	W
R	S	T	U	V	W	R	S	T	U	V	W	R	S	T	U	V	W	R	S	T	U	V	W	X
S	T	U	V	W	X	S	T	U	V	W	X	S	T	U	V	W	X	S	T	U	V	W	X	Y
T	U	V	W	X	Y	T	U	V	W	X	Y	T	U	V	W	X	Y	T	U	V	W	X	Y	Z
U	V	W	X	Y	Z	U	V	W	X	Y	Z	U	V	W	X	Y	Z	U	V	W	X	Y	Z	AA
V	W	X	Y	Z	AA	V	W	X	Y	Z	AA	V	W	X	Y	Z	AA	V	W	X	Y	Z	AA	AB
W	X	Y	Z	AA	AB	W	X	Y	Z	AA	AB	W	X	Y	Z	AA	AB	W	X	Y	Z	AA	AB	AC
X	Y	Z	AA	AB	AC	X	Y	Z	AA	AB	AC	X	Y	Z	AA	AB	AC	X	Y	Z	AA	AB	AC	AD
Y	Z	AA	AB	AC	AD	Y	Z	AA	AB	AC	AD	Y	Z	AA	AB	AC	AD	Y	Z	AA	AB	AC	AD	AE
Z	AA	AB	AC	AD	AE	Z	AA	AB	AC	AD	AE	Z	AA	AB	AC	AD	AE	Z	AA	AB	AC	AD	AE	AF
AA	AB	AC	AD	AE	AF	AA	AB	AC	AD	AE	AF	AA	AB	AC	AD	AE	AF	AA	AB	AC	AD	AE	AF	AG
AB	AC	AD	AE	AF	AG	AB	AC	AD	AE	AF	AG	AB	AC	AD	AE	AF	AG	AB	AC	AD	AE	AF	AG	AH
AC	AD	AE	AF	AG	AH	AC	AD	AE	AF	AG	AH	AC	AD	AE	AF	AG	AH	AC	AD	AE	AF	AG	AH	AI
AD	AE	AF	AG	AH	AI	AD	AE	AF	AG	AH	AI	AD	AE	AF	AG	AH	AI	AD	AE	AF	AG	AH	AI	AJ
AE	AF	AG	AH	AI	AJ	AE	AF	AG	AH	AI	AJ	AE	AF	AG	AH	AI	AJ	AE	AF	AG	AH	AI	AJ	AK
AF	AG	AH	AI	AJ	AK	AF	AG	AH	AI	AJ	AK	AF	AG	AH	AI	AJ	AK	AF	AG	AH	AI	AJ	AK	AL
AG	AH	AI	AJ	AK	AL	AG	AH	AI	AJ	AK	AL	AG	AH	AI	AJ	AK	AL	AG	AH	AI	AJ	AK	AL	AM
AH	AI	AJ	AK	AL	AM	AH	AI	AJ	AK	AL	AM	AH	AI	AJ	AK	AL	AM	AH	AI	AJ	AK	AL	AM	AN
AI	AJ	AK	AL	AM	AN	AI	AJ	AK	AL	AM	AN	AI	AJ	AK	AL	AM	AN	AI	AJ	AK	AL	AM	AN	AO
AJ	AK	AL	AM	AN	AO	AJ	AK	AL	AM	AN	AO	AJ	AK	AL	AM	AN	AO	AJ	AK	AL	AM	AN	AO	AP
AK	AL	AM	AN	AO	AP	AK	AL	AM	AN	AO	AP	AK	AL	AM	AN	AO	AP	AK	AL	AM	AN	AO	AP	AQ
AL	AM	AN	AO	AP	AQ	AL	AM	AN	AO	AP	AQ	AL	AM	AN	AO	AP	AQ	AL	AM	AN	AO	AP	AQ	AR
AM	AN	AO	AP	AQ	AR	AM	AN	AO	AP	AQ	AR	AM	AN	AO	AP	AQ	AR	AM	AN	AO	AP	AQ	AR	AS
AN	AO	AP	AQ	AR	AS	AN	AO	AP	AQ	AR	AS	AN	AO	AP	AQ	AR	AS	AN	AO	AP	AQ	AR	AS	AT
AO	AP	AQ	AR	AS	AT	AO	AP	AQ	AR	AS	AT	AO	AP	AQ	AR	AS	AT	AO	AP	AQ	AR	AS	AT	AU
AP	AQ	AR	AS	AT	AU	AP	AQ	AR	AS	AT	AU	AP	AQ	AR	AS	AT	AU	AP	AQ	AR	AS	AT	AU	AV
AQ	AR	AS	AT	AU	AV	AQ	AR	AS	AT	AU	AV	AQ	AR	AS	AT	AU	AV	AQ	AR	AS	AT	AU	AV	AW
AR	AS	AT	AU	AV	AW	AR	AS	AT	AU	AV	AW	AR	AS	AT	AU	AV	AW	AR	AS	AT	AU	AV	AW	AX
AS	AT	AU	AV	AW	AX	AS	AT	AU	AV	AW	AX	AS	AT	AU	AV	AW	AX	AS	AT	AU	AV	AW	AX	AY
AT	AU	AV	AW	AX	AY	AT	AU	AV	AW	AX	AY	AT	AU	AV	AW	AX	AY	AT	AU	AV	AW	AX	AY	AZ
AU	AV	AW	AX	AY	AZ	AU	AV	AW	AX	AY	AZ	AU	AV	AW	AX	AY	AZ	AU	AV	AW	AX	AY	AZ	BA
AV	AW	AX	AY	AZ	BA	AV	AW	AX	AY	AZ	BA	AV	AW	AX	AY	AZ	BA	AV	AW	AX	AY	AZ	BA	BB
AW	AX	AY	AZ	BA	BB	AW	AX	AY	AZ	BA	BB	AW	AX	AY	AZ	BA	BB	AW	AX	AY	AZ	BA	BB	BC
AX	AY	AZ	BA	BB	BC	AX	AY	AZ	BA	BB	BC	AX	AY	AZ	BA	BB	BC	AX	AY	AZ	BA	BB	BC	BD
AY	AZ	BA	BB	BC	BD	AY	AZ	BA	BB	BC	BD	AY	AZ	BA	BB	BC	BD	AY	AZ	BA	BB	BC	BD	BE
AZ	BA	BB	BC	BD	BE	AZ	BA	BB	BC	BD	BE	AZ	BA	BB	BC	BD	BE	AZ	BA	BB	BC	BD	BE	BF
BA	BB	BC	BD	BE	BF	BA	BB	BC	BD	BE	BF	BA	BB	BC	BD	BE	BF	BA	BB	BC	BD	BE	BF	BG
BB	BC	BD	BE	BF	BG	BB	BC	BD	BE	BF	BG	BB	BC	BD	BE	BF	BG	BB	BC	BD	BE	BF	BG	BH
BC	BD	BE	BF	BG	BH	BC	BD	BE	BF	BG	BH	BC	BD	BE	BF	BG	BH	BC	BD	BE	BF	BG	BH	BI
BD	BE	BF	BG	BH	BI	BD	BE	BF	BG	BH	BI	BD	BE	BF	BG	BH	BI	BD	BE	BF	BG	BH	BI	BJ
BE	BF	BG	BH	BI	BJ	BE	BF	BG	BH	BI	BJ	BE	BF	BG	BH	BI	BJ	BE	BF	BG	BH	BI	BJ	BK
BF	BG	BH	BI	BJ	BK	BF	BG	BH	BI	BJ	BK	BF	BG	BH	BI	BJ	BK	BF	BG	BH	BI	BJ	BK	BL
BG	BH	BI	BJ	BK	BL	BG	BH	BI	BJ	BK	BL	BG	BH	BI	BJ	BK	BL	BG	BH	BI	BJ	BK	BL	BM
BH	BI	BJ	BK	BL	BM	BH	BI	BJ	BK	BL	BM	BH	BI	BJ	BK	BL	BM	BH	BI	BJ	BK	BL	BM	BN
BI	BJ	BK	BL	BM	BN	BI	BJ	BK	BL	BM	BN	BI	BJ	BK	BL	BM	BN	BI	BJ	BK	BL	BM	BN	BO
BJ	BK	BL	BM	BN	BO	BJ	BK	BL	BM	BN	BO	BJ	BK	BL	BM	BN	BO	BJ	BK	BL	BM	BN	BO	BP
BK	BL	BM	BN	BO	BP	BK	BL	BM	BN	BO	BP	BK	BL	BM	BN	BO	BP	BK	BL	BM	BN	BO	BP	BQ
BL	BM	BN	BO	BP	BQ	BL	BM	BN	BO	BP	BQ	BL	BM	BN	BO	BP	BQ	BL	BM	BN	BO	BP	BQ	BR
BM	BN	BO	BP	BQ	BR	BM	BN	BO	BP	BQ	BR	BM	BN	BO	BP	BQ	BR	BM	BN	BO	BP	BQ	BR	BS
BN	BO	BP	BQ	BR	BS	BN	BO	BP	BQ	BR	BS	BN	BO	BP	BQ	BR	BS	BN	BO	BP	BQ	BR	BS	BT
BO	BP	BQ	BR	BS	BT	BO	BP	BQ	BR	BS	BT	BO	BP	BQ	BR	BS	BT	BO	BP	BQ	BR	BS	BT	BU
BP	BQ	BR	BS	BT	BU	BP	BQ	BR	BS	BT	BU	BP	BQ	BR	BS	BT	BU	BP	BQ	BR	BS	BT	BU	BV
BQ	BR	BS	BT	BU	BV	BQ	BR	BS	BT	BU	BV	BQ	BR	BS	BT	BU	BV	BQ	BR	BS	BT	BU	BV	BW
BR	BS	BT	BU	BV	BW	BR	BS	BT	BU	BV	BW	BR	BS	BT	BU	BV	BW	BR	BS	BT	BU	BV	BW	BX
BS	BT	BU	BV	BW	BX	BS	BT	BU	BV	BW	BX	BS	BT	BU	BV	BW	BX	BS	BT	BU	BV	BW	BX	BY
BT	BU	BV	BW	BX	BY	BT	BU	BV	BW	BX	BY	BT	BU	BV	BW	BX	BY	BT	BU	BV	BW	BX	BY	BZ
BU	BV	BW	BX	BY	BZ	BU	BV	BW	BX	BY	BZ	BU	BV	BW	BX	BY	BZ	BU	BV	BW	BX	BY	BZ	CA
BV	BW	BX	BY	BZ	CA	BV	BW	BX	BY	BZ	CA	BV	BW	BX	BY	BZ	CA	BV	BW	BX	BY	BZ	CA	CB
BW	BX	BY	BZ	CA	CB	BW	BX	BY	BZ	CA	CB	BW	BX	BY	BZ	CA	CB	BW	BX	BY	BZ	CA	CB	CC
BX	BY	BZ	CA	CB	CC	BX	BY	BZ	CA	CB	CC	BX	BY	BZ	CA	CB	CC	BX	BY	BZ	CA	CB	CC	CD
BY	BZ	CA	CB	CC	CD	BY	BZ	CA	CB	CC	CD	BY	BZ	CA	CB	CC	CD	BY	BZ	CA	CB	CC	CD	CE
BZ	CA	CB	CC	CD	CE	BZ	CA	CB	CC	CD	CE	BZ	CA	CB	CC	CD	CE	BZ	CA	CB	CC	CD	CE	CF
CA	CB	CC	CD	CE	CF	CA	CB	CC	CD	CE	CF	CA	CB	CC	CD	CE	CF	CA	CB	CC	CD	CE	CF	CG
CB	CC	CD	CE	CF	CG	CB	CC	CD	CE	CF	CG	CB	CC	CD	CE	CF	CG	CB	CC	CD	CE	CF	CG	CH
CC	CD	CE	CF	CG	CH	CC	CD	CE	CF	CG	CH	CC	CD	CE	CF									

$$y_{ijk} = \mu + T_i + A_j + B_k + e_{ijk}$$

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□4) Split Block Design

B1 B5 B2 B5 B4 B1 B2 B4 B3		
B6 B3 B4 B3 B6 B2 B6 B5 B1		
A1 A3 A2		
B4 B2 B1 B4 B2 B5 B3 B2 B5		
B6 B3 B5 B6 B1 B3 B4 B1 B6		
A3 A2 A1		
B2 B5 B6 B4 B3 B6 B4 B3 B2		
B1 B4 B3 B1 B2 B5 B5 B6 B1		
A1 A2 A3		

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B1 B5 B2 B5 B4 B1 B2 B4 B3		
B6 B3 B4 B3 B6 B2 B6 B5 B1		
A1 A3 A2		
B4 B2 B1 B4 B2 B5 B3 B2 B5		
B6 B3 B5 B6 B1 B3 B4 B1 B6		
A3 A2 A1		
B2 B5 B6 B4 B3 B6 B4 B3 B2		
B1 B4 B3 B1 B2 B5 B5 B6 B1		
A1 A2 A3		

$$Y_{ijk} = \mu + \rho_i + \alpha_j + \gamma_{ij} + \beta_k + \alpha\beta_{jk} + \varepsilon_{ijk}$$

$i = 1, 2, \dots, r; j = 1, 2, \dots, a; k = 1, 2, \dots, b$
 ρ_i Block factor (Random)
 α_j Main factor
 γ_{ij} Error for main factor (Random)
 β_k Minor factor
 $\alpha\beta_{jk}$ Interaction between α_j and β_k
 ε_{ijk} Error for minor factor (Random)

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1. The area needed by the treatments of a factor is much more larger than that of another factor. The larger plot is the main plot.
2. The main effect of one factor is more important, and require more accuracy than that of another factor; or the interaction effect is a more important object in research than the main effect; the block in requirement of more accuracy is arranged to be the vice block.
3. According to previous research, the effect of some factor is bigger than that of another factor, the block having more variation is arranged to be the main block.

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□ Strip Block Design

B1	A1B1	A3B1	A2B1
B2	A1B2	A3B2	A2B2

B2	A3B2	A1B2	A2B2
B1	A3B1	A1B1	A2B1

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B1	A1B1	A3B1	A2B1
B2	A1B2	A3B2	A2B2

B2	A3B2	A1B2	A2B2
B1	A3B1	A1B1	A2B1

$$Y_{ijk} = \mu + \rho_i + \alpha_j + \gamma_{ij} + \beta_k + \theta_{ik} + \alpha\beta_{jk} + \varepsilon_{ijk}$$

$i = 1, 2, \dots, r; j = 1, 2, \dots, a; k = 1, 2, \dots, b;$
 ρ_i Block factor (Random)
 γ_{ij} Error for factor A (Random)
 θ_{ik} Error for factor B (Random)
 ε_{ijk} Error for factor A $\times$ B (Random)

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